Hyponatremia — Euvolemic & SIADH (Rule Out Mimics)

1. Five suspects, one is SIADH

WHAT YOU SEE

Banner: FIVE SUSPECTS ONE LOOKS LIKE SIADH

WHAT IT ENCODES

Euvolemic hypotonic hyponatremia has five look-alike causes: glucocorticoid deficiency, hypothyroidism, non-osmotic ADH stimuli (stress/pain/nausea), drugs, and true SIADH. All present euvolemic with concentrated urine (Uosm >100) and urine Na >30, so they are clinically indistinguishable. SIADH is the diagnosis only after the other four are excluded.

BOARD TRAP

WRONG: defaulting to SIADH in any euvolemic hyponatremic patient. The board wants you to first exclude adrenal (cortisol), thyroid (TSH), drugs (SSRI, carbamazepine), and a recent thiazide — each has a specific fix that fluid restriction alone won't provide.

---

2. Euvolemic looks normal

WHAT YOU SEE

Bottom banner: EUVOLEMIC LOOKS NORMAL — RULE OUT MIMICS FIRST

WHAT IT ENCODES

Euvolemic means no signs of volume depletion (no orthostasis, dry mucosa, or azotemia) and no signs of overload (no edema, ascites, JVD). The exam is essentially normal, which is exactly why labs (Uosm, Una, urate, TSH, cortisol) and the medication list drive the diagnosis rather than the physical exam.

BOARD TRAP

WRONG: relying on the physical exam to clinch SIADH. Volume status by exam is notoriously insensitive in the euvolemic range — you must use urine osm, urine Na, uric acid, and hormone testing to separate SIADH from its mimics.

---

3. AVP stuck on

WHAT YOU SEE

Faucet labeled AVP STUCK ON at upper left

WHAT IT ENCODES

The unifying lesion in SIADH is inappropriate, unregulated ADH/AVP secretion despite low serum osmolality. ADH inserts aquaporin-2 channels in the collecting duct, retaining free water, producing concentrated urine and dilutional hyponatremia. Na excretion remains normal (urine Na >30), so the patient is euvolemic rather than edematous.

BOARD TRAP

WRONG: assuming high ADH means the body is volume-depleted. In SIADH the ADH secretion is INAPPROPRIATE (no osmotic or volume stimulus). Giving saline can paradoxically worsen Na in SIADH because the kidney excretes the Na and retains the water.

---

4. SIADH criteria

WHAT YOU SEE

Sign: SIADH SEE OTHER SCENE

WHAT IT ENCODES

Formal SIADH criteria: hypotonic hyponatremia (serum osm <275), inappropriately concentrated urine (Uosm >100), urine Na >30 on normal salt/water intake, clinical euvolemia, and NORMAL thyroid, adrenal, cardiac, hepatic, and renal function. Supportive: low serum uric acid (<4) and low BUN. Classic causes: CNS disease, lung disease (small cell lung cancer, pneumonia), drugs, and pain/post-op states.

BOARD TRAP

WRONG: diagnosing SIADH when urine is dilute (Uosm <100). A dilute urine means ADH is suppressed — that is primary polydipsia or low-solute intake, not SIADH. SIADH requires inappropriately CONCENTRATED urine despite hypotonic plasma.

---

5. Glucocorticoid deficiency

WHAT YOU SEE

Robed figure: GLUCOCORTICOID DEF

WHAT IT ENCODES

Cortisol deficiency — especially SECONDARY/central (pituitary) adrenal insufficiency — causes euvolemic hyponatremia: low cortisol disinhibits CRH/ADH and impairs free-water excretion. In central disease aldosterone is intact, so K is normal (unlike primary Addison). Diagnose with morning cortisol/cosyntropin; treat with hydrocortisone, which corrects the Na briskly.

BOARD TRAP

WRONG: fluid-restricting an isolated glucocorticoid deficiency as if it were SIADH. It mimics SIADH perfectly but needs STEROID replacement; missing it risks adrenal crisis. Always check cortisol before committing to a SIADH label.

---

6. Hypothyroidism

WHAT YOU SEE

Frog figure: HYPOTHYROIDISM, K NORMAL NOT HIGH, CO DOWN GFR DOWN AVP UP

WHAT IT ENCODES

Severe hypothyroidism/myxedema lowers cardiac output and GFR and triggers non-osmotic ADH release, producing euvolemic hyponatremia. Potassium is NORMAL (a clue separating it from adrenal insufficiency, which raises K). Check TSH/free T4; treatment is thyroid hormone replacement, not fluid restriction alone.

BOARD TRAP

WRONG: calling myxedematous hyponatremia SIADH and fluid-restricting. It responds to levothyroxine; SIADH cannot be diagnosed until euthyroidism is confirmed. The normal K (vs high K in Addison) is the discriminator.

---

7. Stress / pain / nausea

WHAT YOU SEE

Figure: STRESS PAIN NAUSEA, NON OSMOTIC AVP

WHAT IT ENCODES

Pain, nausea, severe stress, and the post-operative state are potent NON-osmotic stimuli for ADH release, causing transient euvolemic hyponatremia. This is why post-op patients given hypotonic IV fluids can drop their Na dangerously. Treatment is removing the trigger (analgesia, antiemetics) and avoiding hypotonic fluids.

BOARD TRAP

WRONG: running hypotonic maintenance fluids (e.g., D5 half-normal) in a nauseated post-op patient. Their ADH is already maximally on, so free water is retained and Na falls — use isotonic fluids and treat the pain/nausea. This is a classic cause of fatal hospital-acquired hyponatremia.

---

8. Drugs

WHAT YOU SEE

Figure: DRUGS, AVP RELEASE OR SENSITIZE

WHAT IT ENCODES

Many drugs cause SIADH-like hyponatremia by releasing ADH or sensitizing the kidney to it: SSRIs (esp. in elderly), carbamazepine/oxcarbazepine, cyclophosphamide, vincristine, antipsychotics, NSAIDs, and MDMA/ecstasy. Always scrutinize the medication and recreational-drug history before diagnosing idiopathic SIADH.

BOARD TRAP

WRONG: overlooking the med list and labeling drug-induced hyponatremia as idiopathic SIADH. The fix is stopping the offending agent (e.g., the SSRI or carbamazepine); fluid restriction is only a bridge. MDMA-related hyponatremia in a young patient at a party is a classic vignette.

---

9. Thiazide mimic

WHAT YOU SEE

Capsule figure: THIAZIDE MIMIC, I LOOK LIKE SIADH WAIT 1 TO 2 WK

WHAT IT ENCODES

Thiazides impair urinary dilution while preserving concentrating ability, producing a euvolemic hyponatremia indistinguishable from SIADH — classically in elderly women within 1–2 weeks of starting. Management: STOP the thiazide and recheck Na in 1–2 weeks. Loop diuretics rarely do this because they impair concentration.

BOARD TRAP

WRONG: diagnosing SIADH while the patient is still on a thiazide (HCTZ, chlorthalidone, indapamide). You cannot label SIADH until the drug is discontinued and the Na reassessed — many cases resolve entirely off the thiazide.

---

10. Do not diagnose until thiazide stopped

WHAT YOU SEE

Sign: DO NOT DIAGNOSE SIADH UNTIL THIAZIDE STOPPED

WHAT IT ENCODES

Hard rule for the exam: a current thiazide is an automatic exclusion for SIADH. The thiazide must be stopped and Na rechecked over 1–2 weeks before SIADH can be entertained. This pairs with checking TSH, cortisol, and the rest of the medication list.

BOARD TRAP

WRONG: selecting "SIADH" as the answer when the stem mentions the patient was recently started on hydrochlorothiazide. The intended answer is thiazide-induced hyponatremia — stop the drug first.

---

11. Low urate / bright

WHAT YOU SEE

Coins labeled URATE LOW, BRIGHT, MILD VOLUME EXPANSION

WHAT IT ENCODES

SIADH produces a mild, subclinical volume expansion that triggers natriuresis (urine Na >30) and uricosuria, lowering serum uric acid (<4 mg/dL) and BUN. A low urate supports SIADH; a high urate points to hypovolemia. Despite water retention the patient stays euvolemic because Na is excreted (no edema).

BOARD TRAP

WRONG: expecting edema or weight gain in SIADH. The volume expansion is mild and offset by natriuresis, so the patient is euvolemic, not edematous. A HIGH uric acid argues against SIADH and toward true volume depletion.

---

12. Treat the cause

WHAT YOU SEE

Sign: TREAT THE CAUSE, STOP CULPRIT DRUG

WHAT IT ENCODES

First-line management of euvolemic hyponatremia is cause-directed: stop the offending drug, replace deficient hormone (thyroxine, hydrocortisone), treat the underlying CNS/pulmonary/malignant trigger of SIADH. For true SIADH that persists, add fluid restriction, then oral salt/urea or vaptans.

BOARD TRAP

WRONG: jumping to vaptans/hypertonic saline before removing a reversible cause. If a drug or hormone deficiency is driving it, addressing that fixes the Na — reserve pharmacologic ADH antagonism for refractory true SIADH.

---

13. Fluid restrict if needed

WHAT YOU SEE

Sign: FLUID RESTRICT IF NEEDED

WHAT IT ENCODES

For true SIADH the cornerstone is fluid restriction (often <800–1000 mL/day). If insufficient, escalate to oral salt tablets ± loop diuretic, urea, or vasopressin-receptor antagonists (tolvaptan, conivaptan); demeclocycline is an older second-line option. Correct chronic hyponatremia slowly: no more than ~6–8 mEq/L per 24 hours.

BOARD TRAP

WRONG: giving free water or hypotonic IV fluids in SIADH. Because ADH is on, water is retained and Na falls further. Also WRONG: correcting chronic hyponatremia too fast (>8–10 mEq/L/day) — risks osmotic demyelination syndrome (central pontine myelinolysis).

---

14. Workup labs

WHAT YOU SEE

Checklist: TSH FREE T4, MORNING CORTISOL, COSYNTROPIN, MED LIST, STOP THIAZIDE

WHAT IT ENCODES

The mandatory euvolemic-hyponatremia workup before diagnosing SIADH: TSH/free T4 (thyroid), morning cortisol ± cosyntropin stimulation (adrenal), a full medication review, and discontinuation of any thiazide with reassessment. Supportive labs: serum/urine osm, urine Na, serum uric acid.

BOARD TRAP

WRONG: stopping the workup at urine osmolality and declaring SIADH. Without excluding thyroid and adrenal disease (and a thiazide), the diagnosis is incomplete and you may miss a hormone deficiency that needs replacement rather than fluid restriction.